ICQCM 2026

Science, Causal Inference, And Frankenstein

Evaluating Neighborhood Crime Interventions using Synthetic Control Methods

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https://github.com/YungChun0303/ICQCM

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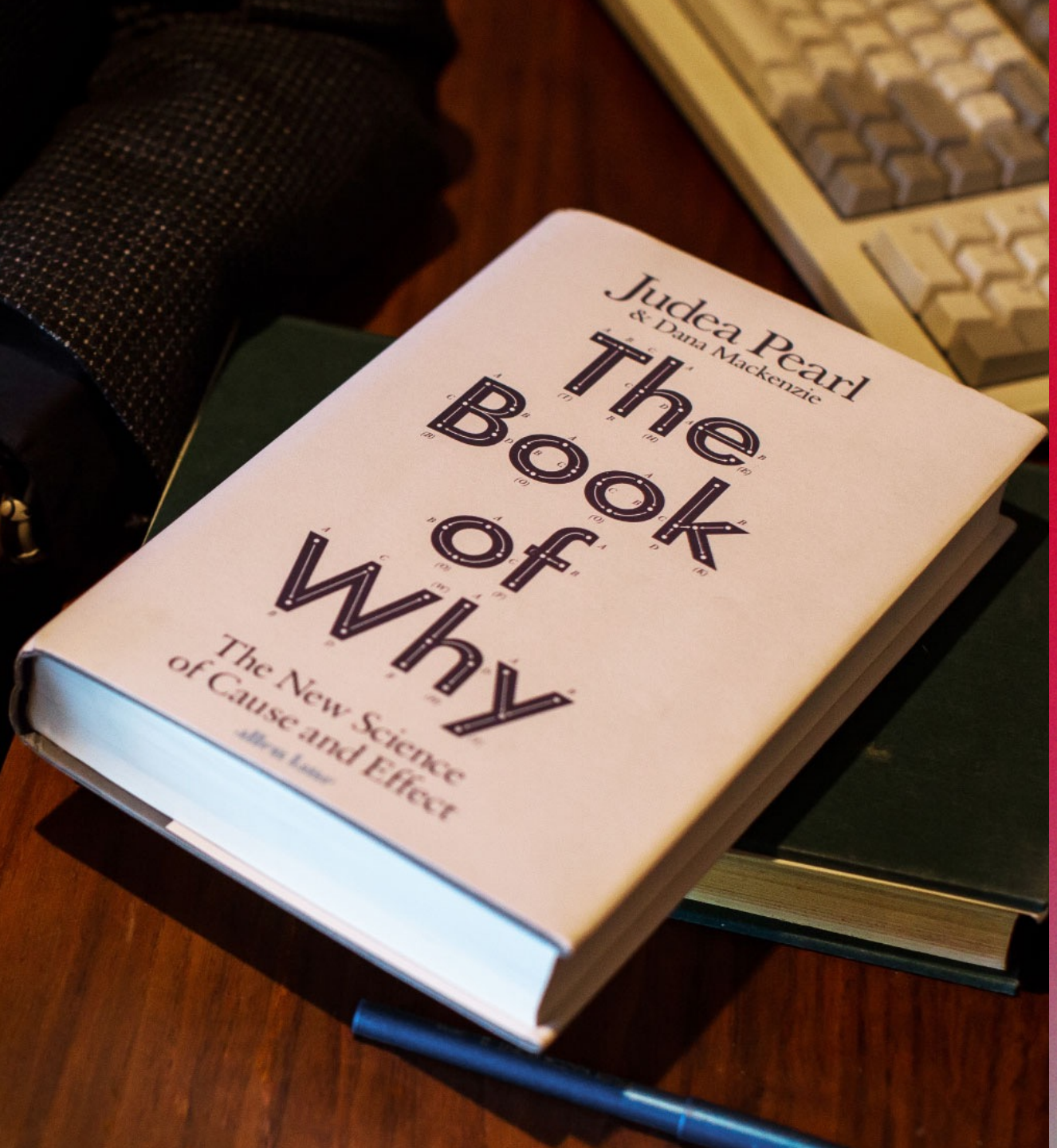
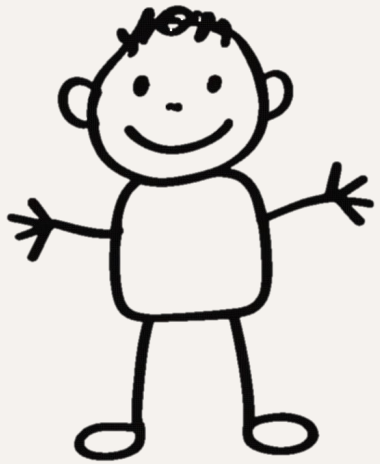

Judea Pearl
& Dana Mackenzie
The Book of Why
The New Science of Cause and Effect

Science?

What is Science?



Science is like being a detective for the whole world

It's all about asking **'Why?' and 'How?'** to find out what makes things happen.

Think of it like a game of **Cause and Effect**—if I push this button, does the light turn on? If it does, I found the cause!

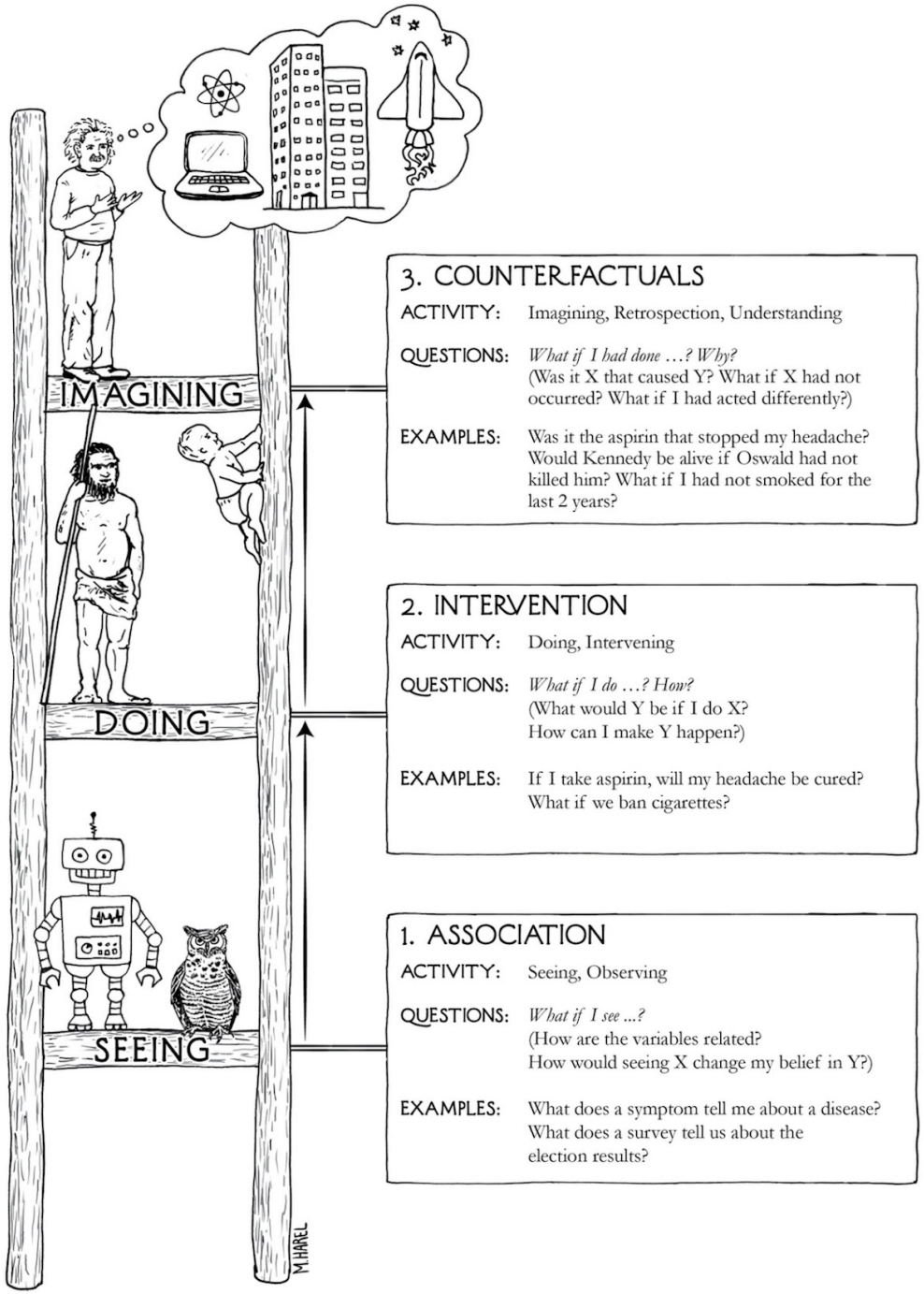
–Nathan Chun, while playing *The Legend of Zelda*

Science...?

Causal Inference

It may be our only hope for surviving the AI invasion.

> Deep learning has indeed achieved amazing things... but it is essentially a very sophisticated form of curve fitting. It works by finding patterns in data, but it does not have a model of the world. **It cannot reason about what would happen if it intervened in the world**, or **what might have happened in an alternative reality**.
>
> – Judea Pearl, *The Book of Why*



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Causal Identification (in Social Science)

This is our (my) bread and butter (and why you are now in New Orleans).

Method	The "Counterfactual" Source	Best Used When...	Example
Randomized Controlled Trials	Random assignment creates a "twin" group automatically.	You have total control and a big budget.	A/B test
Natural Experiment	Nature or policy creates a "random-like" shock (e.g., birth dates, lotteries).	An external event mimics a laboratory setting.	IV, RDD
Quasi-Experiment	A similar group that didn't get the treatment (the "Control").	You have a clear "Before/After" and a "Treatment/Control" group.	DID, 2WFE, OLS, ..., SCM

Quasi-Experiment

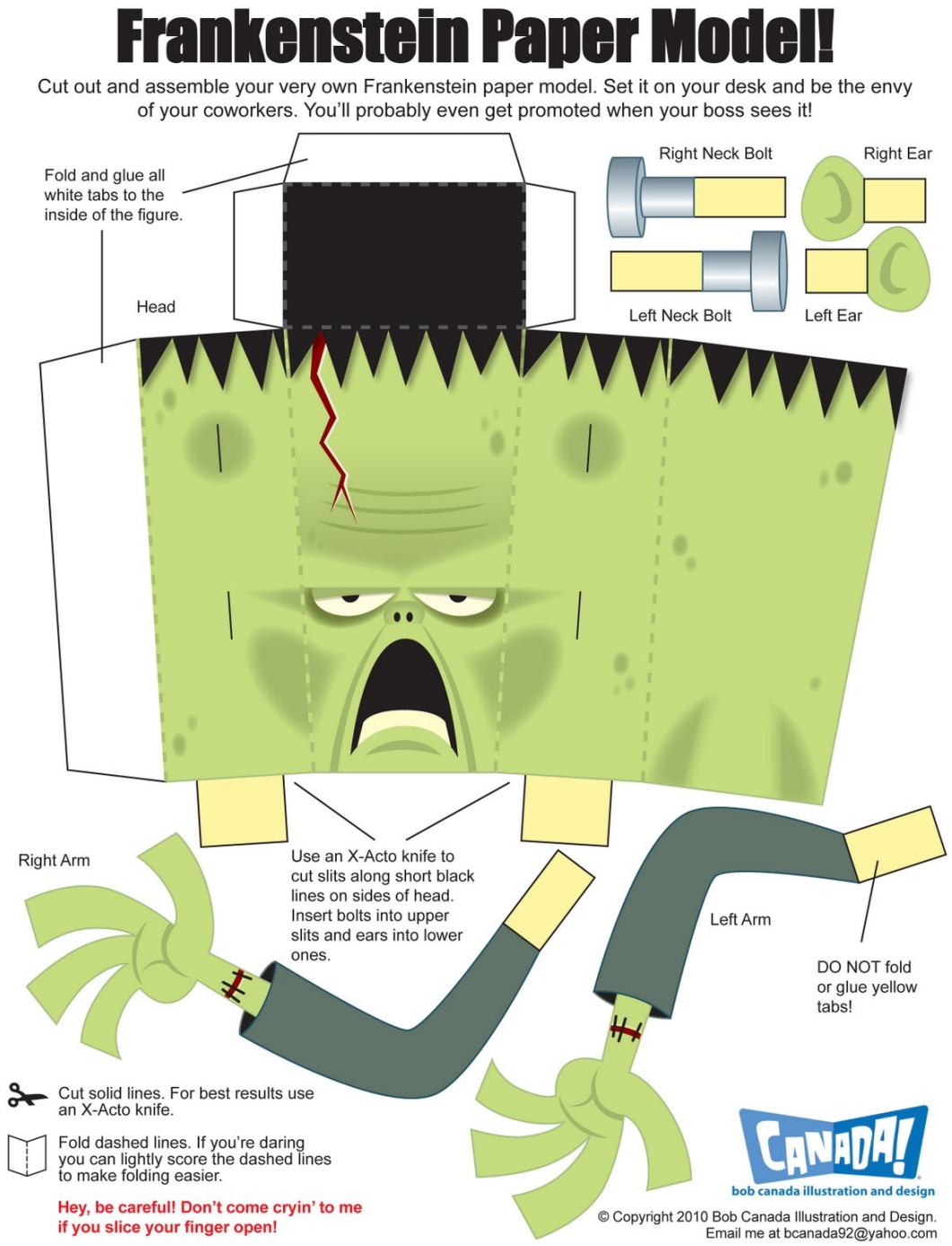
Quasi-Experiments Are Also Costly — But SCM Changes That!

	Conventional Quasi-Experiment	Synthetic Control Method (SCM)
Control Group	**Selected:** You find a "neighbor" or "similar" city that already exists.	**Constructed:** You "build" a twin using parts from many different cities.
Key Assumption	**Parallel Trends:** You assume the control and treatment would have moved the same way.	**Pre-intervention Fit:** You prove the "Frankenstein" mirrors the treatment group for a long time before the shock.
Unit Level	Works well for large groups (e.g., all people in a state).	Designed specifically for **Comparative Case Study.**

Science...?

Doppelganger and the Frankenstein

Meeting Doppelganger is hard; Making a Frankenstein is less hard.



Synthetic Control Model (SCM)

The Problem of the "Unique" Unit

Why does Difference-in-Differences (DiD) often fail in neighborhood research?

The Comparative Case Study Challenge

- We often study unique, place-based interventions (e.g., the CNI project in a metro area)

The DID Failure

- Standard quasi-experiments require a "control" unit that shares parallel trends.
- In reality, no single neighborhood in a metro area is a "perfect" match for Sun Valley's pre-treatment trajectory.

The SCM Pivot

- If you can't find a peer, you must *manufacture* one. We move from searching for a control to *constructing* one.

The Identification Strategy

You may copy and paste this slide for your papers.

The Premises:

- **The Donor Pool (J):** A set of untreated unites that could potentially resemble the treated units.
- **The Synthetic Unit:** A weighted Average of these donors.

$$\mathbb{W} = \left(w_2, \dots, w_{J-1}\right)'$$

where weights are non-negative and sum to 1.

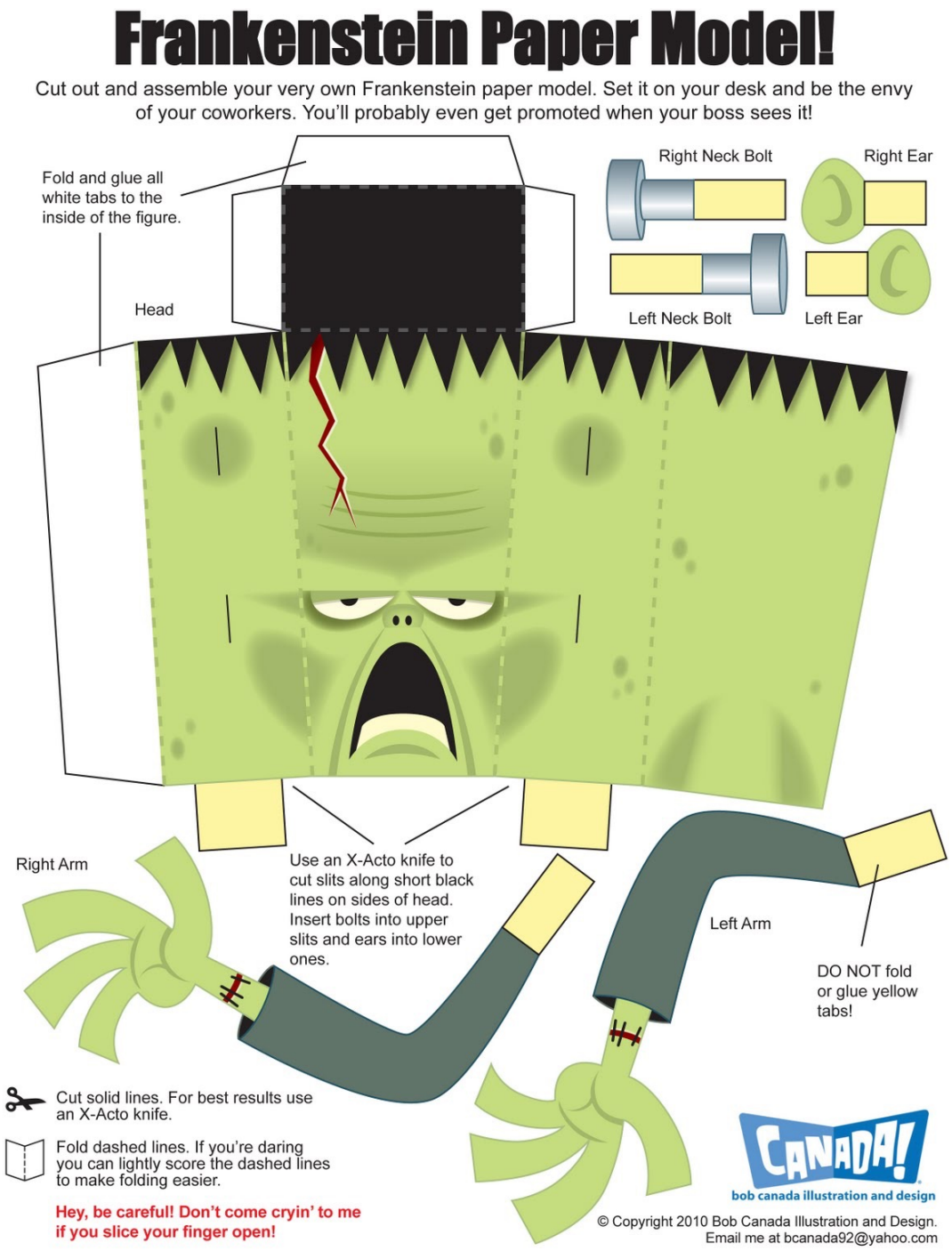
The objective

- We choose weights ($\mathbb{W}$) such that the synthetic unit's pre-treatment outcomes ($Y_{\mathbb{W}*J,0}$) and characteristics ($\mathbb{X}_{\mathbb{W}*J,0}$) nearly perfectly match the treated unit ($Y_{i,0}$, $\mathbb{X}_{i,0}$)

Synthetic Control

Wait, what does that mean?

We are making a Frankenstein!



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Wait, what does that mean?

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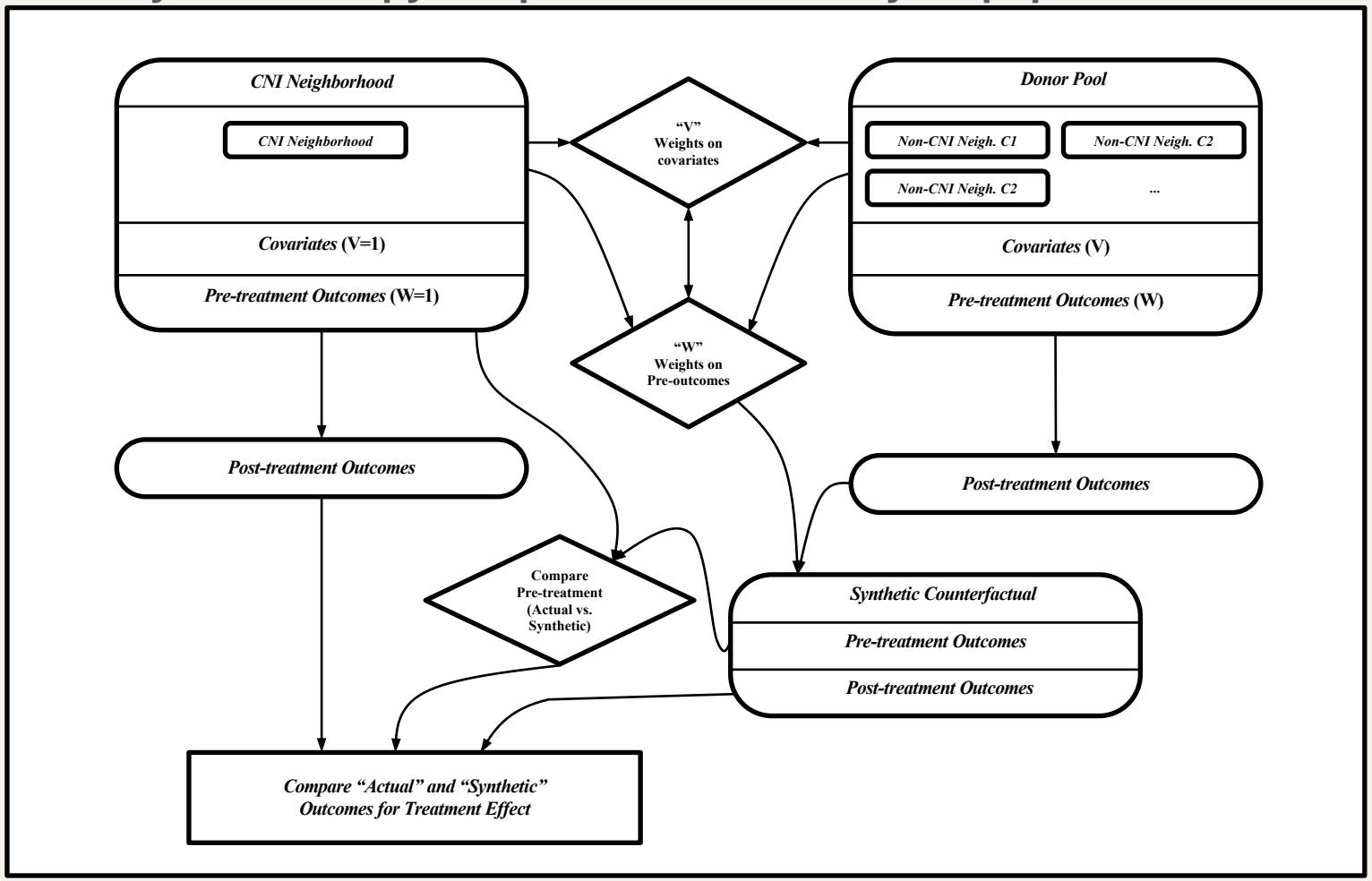

Chun et al. (2022)

The Identification Strategy

You may copy and paste this slide for your papers.

- **The Model:** Let Y_{it} be the outcome. We want to estimate the effect τ_{1t} for the treated unit ($i = 1$) at time $t > T_0$:

$$\tau_{1t} = Y_{1t} - \sum_{j=2}^{J+1} w_j * Y_{jt}$$

- **The Weights ($\mathbb{W}$):** Found by minimizing the distance between treated and synthetic predictors:

$$\|\mathbb{X}_1 - \mathbb{X}_0 W\| = \sqrt{(\mathbb{X}_1 - \mathbb{X}_0 \mathbb{W})' \mathbb{V} (\mathbb{X}_1 - \mathbb{X}_0 \mathbb{W})}$$

  - **The $\mathbb{V}$ Matrix:** A diagonal matrix representing the "importance" of each predictor.
    - This is the "brain" of the Frankenstein—deciding which variables (e.g., prior crime rates vs. poverty levels) matter most for the match.

Placebo Tests

Wait, given one-to-one comparison, then how do we estimate confidence intervals (and STARs)?

In-Time Placebo:

Moving the treatment date back in time. If SCM shows an "effect" before the grant actually happened, the model is broken.

In-Space Placebo (Permutation)***

Re-running the model for every neighborhood in the donor pool as if they received the grant.

- The P-Value: We calculate the ratio of post-treatment error to pre-treatment error (RMSPE). If the real treated is larger than almost all fake treatments, the result is statistically significant.

Does CNI grant reduce crime rate in the neighborhood?

BJCI Grants in Sun Valley Neighborhood in Denver, CO

A wicked problem

Crime as a Wicked Problem:

- ***Blame Avoidance*:** Under salient pressure, agencies prioritize plausible outputs (demonstrating they "tried") over actual outcomes (solving the problem).
- ***Bureaucratic Circumvention:*** Decoupling daily tasks from the ultimate policy goal.

Research Question:

- **RQ1:** Did the CNI grant lead to a measurable change in **policing behavior** in the Sun Valley neighborhood?
- **RQ2:** Did this change in behavior translate into a reduction in **actual crime counts**?

Project Overview — BCJI in Denver

The Program:

- The neighborhood received a $1 million+ federal grant through the Byrne Criminal Justice Innovation (BCJI) program.

The Target:

- Sun Valley neighborhood, Denver, CO—a high-poverty area identified for place-based, data-driven crime mitigation.

The Sun Valley Context:

- A two-year implementation window (2017–2018) characterized by intense scrutiny and the need to demonstrate "progress" to federal stakeholders.

Synthetic Control Model

In R: tidysynth & the Denver CNI Case Study

A Synthetic Control Model in R

An R package for Synthetic Control analysis in a tidy, pipe-friendly workflow

The Core Idea:

- Build a weighted average of untreated 'donor' units that best matches the treated unit before the intervention — this becomes the counterfactual.

Limitation of Diff-in-Diff:

- SCM is better suited when there is only one (or few) treated units — DiD requires many treated units for variance estimation.

Key Assumption:

- Pre-treatment parallel trends: the synthetic control must closely track the treated unit before treatment starts, justifying it as a counterfactual.
- Inference via Placebos: Statistical significance is assessed by 'in-space' placebo tests — running SCM on every donor as if it were treated.

Data Overview

Denver's Crime & Neighborhood Initiative (CNI)

Data

Timeline

- De facto launch: 2016 Q3
- Project end: 2018 Q3
- Balanced panel: 44 quarters (2011 Q1 – 2021 Q4)

Treatment

- Targeted one census tract (FIPS: 8031000800)
- Will exclude adjacent tracts due to potential spillover effects

Outcomes of interests

- Crime panel: all crimes, property, violent, drug
- Stop panel: all stops, no-action, citations, arrests

Covariates:

- median income, race, population, population density

What We Will Do: The analysis Pipeline

01 Load & Prepare

Import balanced crime and stop panels. Filter adjacent tracts and rename variables.

02 Descriptive Trends

Plot raw CNI vs. donor averages to visually inspect pre-treatment parallel trends.

03 Specify SCM

Use tidysynth to declare unit/time structure, select predictors (demographics + pre-treatment outcome mean).

04 Estimate Weights

Solve constrained optimisation to find donor weights minimising pre-treatment prediction error.

05 Compute Effects

Compare observed vs. synthetic trajectories post-2016 Q3. Report average gap and % change for 8 outcomes.

06 Placebo Tests

Run in-space placebos on all donors. Compute MSPE ratios and Fisher p-values to assess significance.

tidysynth

An R package for Synthetic Control analysis in a tidy, pipe-friendly workflow

`tidysynth` is:

- an R package that implements the Synthetic Control Method (SCM) using a tidy, pipe-friendly (%>%) workflow.
- designed around a sequential function chain — each function does exactly one job, and you build the model step by step.

The object returned at each stage is enriched in place, so you can inspect intermediate results at any point.

The Four Core Functions

The entire SCM pipeline consists of exactly four functions called in order:

```
data %>%
    synthetic_control(...) %>%          # Step 1: Declare panel structure
    generate_predictor(...) %>%         # Step 2: Specify matching predictors
    generate_weights(...) %>%           # Step 3: Solve for donor weights
    generate_control()                  # Step 4: Build counterfactual
```

Step 1. Declare the panel structure

This registers the essential dimensions of your data. Nothing is estimated yet — this is purely declarative.

```
synthetic_control(
  outcome          = allstop,          # The variable you want to study
  unit             = FIPS,             # Column identifying each unit
  time             = qtr,              # Column identifying each time period
  i_unit           = 8031000800,       # Which unit received treatment
  i_time           = 2016.5,           # When treatment started
  generate_placebos = FALSE            # Whether to run placebo models
)
```

Step 2a. Specify what to match on

This tells which variables to use when finding donor weights, and over which time window to compute them.

```
generate_predictor(
    time_window = 2015,                # Period over which the predictor is computed
    med_inc_k   = med_inc / 1000,      # covariate 1
    pct_color   = pct_color,           # covariate 2
    pop         = pop,                 # covariate 3
    pop_density = pop_density          # covariate 4
    …
)
```

Step 2b. Specify what to match on

You can call generate_predictor() multiple times to add different sets of predictors from different time.

```
generate_predictor(
    time_window = seq(2011, 2015.75, 0.25),        # every quarter, 2011–2015
    outcome_pre = mean(allstop, na.rm = TRUE)
)
```

Step 3. Solve for donor weights

This runs the constrained optimization at the heart of SCM.

```
generate_weights(
    optimization_window = seq(2011, 2016, 0.25)  # The time window used for fitting
)
```

Step 4. Build the counterfactual

To produce the synthetic counterfactual trajectory (No arguments needed).

```
stop_panel %>%
  synthetic_control(
    outcome  = allstop,
    unit     = FIPS,
    time     = qtr,
    i_unit   = FIPS_CNI,
    i_time   = 2016.5
  ) %>%
  generate_predictor(...) %>%
  generate_weights(
    optimization_window =
      seq(2011, 2015.75)
  ) %>%
  generate_control()
```

Key Outputs We Will Produce

1. **Trend Plots**
   Observed vs. synthetic trajectories for crime/stops outcomes

2. **Balance Table**
   Predictor balance check: CNI tract vs. synthetic control

3. **Donor Weight Table**
   Which tracts contribute to the synthetic, and how much

4. **Effect Summary Table**
   Avg gap & % change for all outcomes post-2016 Q3

5. **Placebo Plots**
   In-space placebo overlays showing CNI vs. donor noise

6. **Significance Table**
   MSPE ratio rankings and Fisher p-values per outcome

tidysynth workflow

```
stop_panel %>%
  synthetic_control(
    outcome   = allstop,
    unit      = FIPS,
    time      = qtr,
    i_unit    = FIPS_CNI,
    i_time    = 2016.5
  ) %>%
  generate_predictor(...) %>%
  generate_weights(
    optimization_window =
      seq(2011, 2015.75)
  ) %>%
  generate_control()
```

Do I really need to know all of this?

The Next Program Language is… English

Please create an R Markdown document for a Synthetic Control Method (SCM) case study using the tidysynth package in R. Here are the exact specifications:

Study context: Denver's Choice Neighborhood Initiative (CNI), evaluating effects on crime and police stops at the census tract level.

Data structure:

- Two balanced quarterly panels: denver_crime_qtr.csv and denver_stop_qtr.csv
- Panel covers 2011 Q1 – 2021 Q4 (44 quarters), time encoded as numeric yearqtr (e.g., 2011.0, 2011.25, 2011.5, 2011.75)
- Key columns: FIPS (census tract ID), qtr (time), group (1=treated, 2=adjacent, 3=donors), and outcome variables
- Treated unit FIPS: 8031000800; treatment start: 2016.5 (2016 Q3)
- Filter out group == 2 (adjacent tracts) before analysis

Outcomes to model:

- Crime panel: Allcrime, Property, Violence, Drug
- Stop panel: allstop, no_action, n_citation (rename from citation), arrest

SCM predictors (all measured pre-treatment):

- At year 2015: med_inc / 1000 (median income in $1,000s), pct_color (% non-white), pop, pop_density
- Pre-treatment outcome mean over seq(2011, 2015.75, 0.25)
- Optimization window: seq(2011, 2015.75, 0.25)

Required sections:

1. Setup (libraries: tidyverse, tidysynth, knitr, kableExtra, here)
2. Load & prepare data
3. Descriptive trend plots (treated vs. donor average, with vertical dashed lines at 2014.0, 2016.5, 2018.5)
4. Step-by-step SCM walkthrough using police stops as the running example (Steps 1–5: declare panel, specify predictors, estimate weights, build synthetic control, inspect fit)
5. A reusable run_synth() function and plot_synth_trends() function
6. SCM results for all 8 outcomes (4 crime + 4 stop), each with a trend plot
7. Treatment effect summary tables (gap = observed − synthetic, post-treatment period)
8. In-space placebo tests for no_action and Drug crime, with plot_placebos() and MSPE ratio tables
9. Save all plots to an outputs/ folder using ggsave()

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